

Problem

Show that the I^2R losses will be higher for a 120-V appliance than for a 240-V appliance if both have the same power rating.

Step-by-step solution

Step 1 of 2

If both appliances have the same power rating P ,

$$I = \frac{P}{V_s}$$

For the 120-V appliance,

$$I_1 = \frac{P}{120}$$

For the 240-V appliance,

$$I_2 = \frac{P}{240}$$

$$\text{Power Loss} = I^2 R = \begin{cases} \frac{P^2 R}{120^2} & \text{For the 120-V appliance} \\ \frac{P^2 R}{240^2} & \text{For the 240-V appliance} \end{cases}$$

Step 2 of 2

Since,

$$\frac{1}{120} > \frac{1}{240^2}$$

The losses in the 120-V appliances are higher.